Novelty & Differentiation Report

DSCI-41: Traffic Volume Estimation and Short-Term Congestion Forecasting from Roadside Video

Project ID: DSCI-41 | Domain: Visual Intelligence Systems | Specialization: AI Systems for Visual Intelligence

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1. Introduction

This report presents a detailed comparison between our proposed capstone project and the existing body of research, including the faculty-assigned reference — the Zindi Barbados Traffic Analysis Challenge (Nov 2025 – Jan 2026). The purpose is to clearly document what has already been done, what gaps remain, and what our project will do differently to contribute new knowledge.

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2. What Existing Research Has Done

2.1 Object Detection for Traffic (Published Papers)

Existing research papers on vehicle detection from CCTV/roadside cameras typically follow this pipeline:

Input (Video Frame) → YOLO Detection → Bounding Boxes → Count

Paper Category	What They Do	What They DON'T Do
YOLOv8 vehicle counting papers (2023-2024)	Detect and count vehicles per frame using virtual counting lines	They stop at counting — no forecasting of future congestion
DeepSORT / ByteTrack tracking papers	Assign unique IDs to vehicles across frames to avoid double-counting	They test only on clear weather datasets (no rain, night, fog)
Perspective correction papers (Homography)	Use a 3×3 matrix to convert oblique camera angles to bird's-eye view	They correct perspective but never measure how much the correction improves downstream counting accuracy
Adverse weather detection papers (2024)	Test YOLOv8 on rain/fog/night images from ACDC or BDD100K datasets	They evaluate on static images only, never on continuous video streams where tracking also degrades

Gap identified: Every paper treats detection, tracking, perspective correction, weather robustness, and forecasting as isolated problems. No published work connects them into a single end-to-end pipeline and measures how errors propagate from one stage to the next.

2.2 Traffic Forecasting (Published Papers)

Existing forecasting papers predict future traffic volume using time-series models:

Paper Category	What They Do	What They DON'T Do
LSTM / Bi-LSTM traffic prediction papers	Predict traffic volume 15-60 minutes ahead using historical count data	Their input data comes from inductive loop sensors or GPS traces — NOT from a camera/vision system
CNN-LSTM hybrid papers	Combine spatial (CNN) and temporal (LSTM) features for better accuracy	They assume the input count data is perfect (no detection errors)
XGBoost traffic papers	Use gradient boosting with tabular features (time, day, weather)	The "weather" feature is manually entered — not automatically detected from the video
Graph Neural Network papers	Model spatial dependencies between multiple road segments	Require a graph of the road network — over-engineered for a single camera location

Gap identified: Every forecasting paper assumes clean, sensor-grade input data. None of them study what happens when the input comes from a noisy, error-prone computer vision system where rain causes missed detections and night causes false positives.

2.3 The Reference Paper: Zindi Barbados Traffic Analysis Challenge

The faculty-assigned reference is a data science competition, not a traditional research paper. Here is exactly what the competition did and did not do:

What the Zindi competition DID:

- Provided 4 video streams from a roundabout in Barbados
- Required participants to predict congestion levels (free flowing / light / moderate / heavy delay) 5 minutes into the future
- Used Macro-F1 (70%) + Accuracy (30%) as the evaluation metric
- Winners used feature engineering from video (object detection → counting → tabular features → classification)

What the Zindi competition DID NOT do:

Limitation	Explanation
No adverse weather testing	All Barbados videos were recorded in clear, tropical weather. There is zero rain, fog, or night-time footage in the dataset
No perspective handling	The cameras were fixed at specific angles. Winners did not need to handle oblique views or cross-camera generalisation
No real-time deployment	It was an offline competition — participants processed pre-recorded video. No live camera feed was involved
No uncertainty quantification	All predictions were single-class labels. No confidence intervals or "I don't know" abstention mechanism existed
No error propagation study	Winners did not analyse how detection errors in the vision system affected the quality of the congestion forecast

Single location only	The entire competition was limited to one roundabout in Barbados. No generalisation to other roads or countries was attempted
Backpropagation banned	The competition rules prohibited end-to-end neural network training, forcing participants to use only classical feature engineering

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3. What OUR Project Does Differently (Novel Contributions)

Our project addresses five specific gaps that neither existing research papers nor the Zindi reference competition have solved. Each contribution is designed to be independently testable and measurable — as required by the KL University capstone evaluation framework (O3 contribution isolation).

3.1 Contribution 1: End-to-End Vision-to-Forecast Pipeline with Error Propagation Analysis

What exists: Detection papers stop at counting. Forecasting papers start with clean sensor data. Nobody connects the two.

What we do differently: We build a single pipeline where the vision module's output (vehicle counts) feeds directly into the forecasting module. Critically, we measure error propagation — i.e., we quantify how much a 10% detection error during rain translates into forecasting error.

How we measure it:

- Run the forecaster with ground-truth counts (from the Zindi reference) → Record forecast accuracy (Baseline A)
- Run the forecaster with vision-estimated counts (from our YOLO + ByteTrack pipeline) → Record forecast accuracy (Baseline B)
- Calculate the propagation gap = Baseline A – Baseline B
- Report this gap per condition (clear / rain / night)

Why it matters: If the gap is large during rain, it tells traffic authorities that the system's forecasts should not be trusted blindly during bad weather. No prior work provides this information.

3.2 Contribution 2: Condition-Aware Adaptive Preprocessing

What exists: Current systems apply the same preprocessing to every frame regardless of weather. Papers either always apply CLAHE (brightness enhancement) or never apply it.

What we do differently: We build a lightweight condition classifier (a small CNN or even a rule-based model using histogram analysis) that automatically tags each incoming frame as one of: clear | rain | night | fog. Based on this tag, the system applies condition-specific preprocessing:

Detected Condition	Preprocessing Applied

Clear / Daytime	None (raw frame)
Night / Low Light	CLAHE (Contrast Limited Adaptive Histogram Equalization)
Rain	Deraining filter + contrast enhancement
Fog	Dark Channel Prior dehazing

Why it matters: Applying CLAHE to bright daytime frames actually degrades detection accuracy by introducing artifacts. Our adaptive approach avoids this problem. No prior traffic counting paper implements this automatic switching.

3.3 Contribution 3: Confidence-Weighted Soft Counting

What exists: Every existing vehicle counting system uses hard counting — if YOLO detects a vehicle (regardless of confidence), it counts as exactly 1.

What we do differently: We introduce soft counting, where each detection is weighted by its confidence score:

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Hard count:  Car (0.95) = 1,  Car (0.55) = 1,  Total = 2
Soft count:  Car (0.95) = 0.95,  Car (0.55) = 0.55,  Total = 1.50
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Why it matters: During rain or night, YOLO produces many low-confidence detections (it's "guessing"). Soft counting naturally captures this uncertainty and passes it downstream to the forecaster. The forecaster then learns that low soft-counts indicate uncertain input, enabling it to produce wider confidence intervals for its predictions.

How we validate: Compare hard-count-based forecasting vs. soft-count-based forecasting. We expect soft counting to produce:

- Similar accuracy during clear weather (both methods work well)
- Better accuracy during adverse weather (soft counting avoids over/under-counting)

3.4 Contribution 4: Weather-Conditioned Forecasting

What exists: Current LSTM and XGBoost forecasting models take only two types of input: (1) past vehicle counts and (2) time features (hour, day of week). They are completely blind to weather conditions.

What we do differently: We add the automatically detected weather condition (from Contribution 2's classifier) as an explicit input feature to the forecasting model:

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Existing model input:  [count_t-1, count_t-2, ..., hour, day_of_week]
OUR model input:       [count_t-1, count_t-2, ..., hour, day_of_week, weather_tag, avg_confidence]
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Where:

- weather_tag = {0: clear, 1: rain, 2: night, 3: fog} — from condition classifier
- avg_confidence = mean YOLO detection confidence for that time interval — from soft counting

Why it matters: Traffic behaves fundamentally differently during rain (slower speeds, more cautious driving, longer congestion). A model that knows it's raining can predict "congestion will last 30 minutes" instead of the standard "congestion will clear in 15 minutes." The Zindi Barbados competition did not have adverse weather data, so this capability was never developed or tested.

3.5 Contribution 5: Real-Time Deployment on Live Camera Feeds

What exists: The Zindi Barbados competition was entirely offline (pre-recorded videos processed in notebooks). All published papers also evaluate on recorded datasets.

What we do differently: We demonstrate our system running in real-time on:

1. YouTube live traffic camera streams (24/7 feeds from various cities)
2. Our own recorded Indian traffic footage (from a camera near KL University)

We build a Streamlit dashboard that shows:

- Live video feed with bounding boxes around detected vehicles
- Real-time vehicle count (updated every 5 minutes)
- Congestion forecast for the next 15-30 minutes
- Confidence indicator ("System is confident" vs. "Uncertain — adverse conditions detected")

Why it matters: This transforms the project from a research exercise into a deployable prototype. No published paper or Zindi entry has demonstrated a live, end-to-end traffic forecasting system.

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4. Direct Comparison with Specific State-of-the-Art Papers

To further highlight our novelty, here is how our proposed pipeline compares directly to recent, highly-cited research papers published in 2023–2024 on similar topics.

State-of-the-Art Paper	Their Core Approach	What They Missed	How DSCI-41 Solves It
Chen & Chen (2024) "CNN-LSTM Hybrid for Short-Term Traffic Flow Prediction"	High-accuracy forecasting using spatial-temporal CNN-LSTM hybrids.	Relied entirely on clean inductive loop sensor data. Did not address vision-based detection errors.	Uses computer vision directly and feeds soft-counts + weather context into the LSTM, closing the gap between noisy vision and
Zhang et al. (2022) "ByteTrack: Multi-Object Tracking by Associating Every Detection Box"	Best-in-class multi-object tracking for dense traffic using YOLO.	Tested primarily on daytime/clear datasets. Fails silently during heavy rain when detection confidence drops.	Implements weather-aware adaptive preprocessing (CLAHE/derain) before tracking to maintain ByteTrack's accuracy at night and in
Sakaridis et al. (2021) "ACDC: The Adverse Conditions Dataset"	Trained semantic models to detect vehicles in fog, rain, and snow.	Evaluated purely on static image benchmarks. Did not track vehicles or estimate traffic volume.	Applies adverse-weather weights to a continuous video stream, enabling robust counting and flow estimation under harsh conditions.
Zindi Barbados Challenge (2026) "Traffic Analysis Winning Solutions"	Feature engineering (time in roundabout, flow rate) + classification algorithms.	Offline only. Clean tropical weather. Single static roundabout. No error propagation study.	Real-time deployment. Handles adverse weather actively. Implements confidence-weighted counting to quantify uncertainty.

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5. Summary Comparison Table

Feature	Existing Papers	Zindi Barbados (Reference)	OUR Project
Vehicle Detection	YOLO-based	YOLO-based	YOLOv8 (same baseline)
Multi-Object Tracking	DeepSORT / ByteTrack	Used by winners	ByteTrack (same baseline)
Perspective Handling	Homography	Not needed (fixed cameras)	Homography + cross-camera validation
Adverse Weather Robustness	Static images only	All clear weather	Video-level testing on rain/night/fog
Adaptive Preprocessing	Same for all frames	Not attempted	Condition-specific (CLAHE/dehaze/derain)
Soft Counting	Hard count only	Hard count only	Confidence-weighted soft counts
Congestion Forecasting	From sensors only	From video features	From vision counts + weather tag
Weather as Forecast Input	Not used	No weather data	Auto-detected weather feeds into forecaster
Error Propagation Study	Not studied	Not studied	Measured per condition
Uncertainty / Confidence	Single prediction	Single class label	Prediction + confidence intervals
Real-Time Live Demo	Offline only	Offline competition	Live YouTube + local camera feed
Indian Traffic Testing	Western data only	Barbados only	Indian mixed traffic (autos, bikes, trucks)

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6. How Each Novel Contribution Maps to Project Objectives

Novel Contribution	CP-1 or CP-2	Objective
End-to-End Pipeline + Error Propagation	CP-2	O4 (Integration) + O5 (Validation)
Adaptive Preprocessing	CP-2	O3 (Differentiating Contribution)
Confidence-Weighted Soft Counting	CP-1 (design) + CP-2 (implement)	O2 (Reference) + O3 (Contribution)
Weather-Conditioned Forecasting	CP-2	O4 (Integration)
Real-Time Live Demo	CP-2	O4 (Integration) + O5 (Final Demo)

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7. Conclusion

Our project builds upon the Zindi Barbados Traffic Analysis Challenge reference and extends it in five measurable ways. While the reference competition solved the problem of predicting roundabout congestion from clear-weather video, our project tackles the harder, real-world problem of maintaining accuracy under adverse conditions, quantifying uncertainty, and deploying the system in real-time on live camera feeds.

The combination of adaptive preprocessing, confidence-weighted counting, and weather-conditioned forecasting represents a genuine technical contribution that has not been published in any prior research paper on traffic volume estimation from roadside video.

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8. References

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